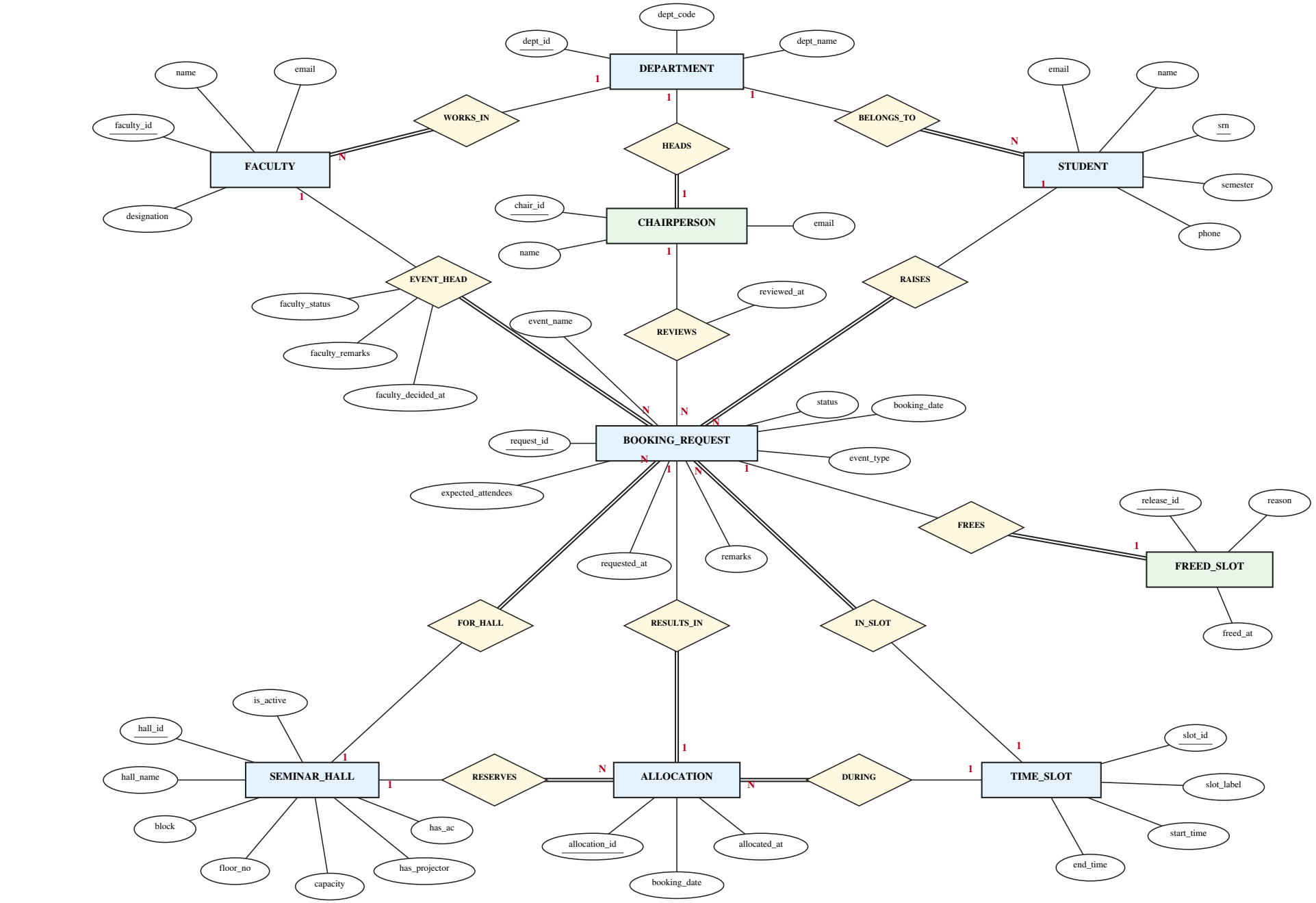
University Seminar Hall Booking System - ER Diagram (Chen notation)



Green = added for our college's rules (chairperson approval, freed slots). Ovals on a diamond = attributes of that relationship

Relational Schema (9 tables)

Primary keys underlined, foreign keys in italics.

DEPARTMENTS

dept_id	dept_code	dept_name

CHAIRPERSONS new: our college rules

chair_id	name	email	*dept_id*

- dept_id → DEPARTMENTS (UNIQUE: one chairperson per department)

STUDENTS

srn	name	email	phone	semester	*dept_id*

- dept_id → DEPARTMENTS

FACULTY

faculty_id	name	email	designation	*dept_id*

- dept_id → DEPARTMENTS

SEMINAR_HALLS

hall_id	hall_name	block	floor_no	capacity	has_projector	has_ac	is_active

TIME_SLOTS

slot_id	slot_label	start_time	end_time

- CHECK start_time ≥ 09:00 AND end_time ≤ 16:00
- CHECK end_time − start_time ≤ 2 hours
- 30 minute gap between slots (trigger trg_slot_gap_insert / _update)

BOOKING_REQUESTS

request_id	*srn*	*hall_id*	*slot_id*	booking_date	event_name	event_type	expected_attendees	*faculty_id*	faculty_status	faculty_remarks	faculty_decided_at	status	remarks	requested_at	*reviewed_by*	reviewed_at

- srn → STUDENTS
- hall_id → SEMINAR_HALLS
- slot_id → TIME_SLOTS
- faculty_id → FACULTY (the event head, who approves first)
- reviewed_by → CHAIRPERSONS (the final approval; NULL until the chairperson decides)

ALLOCATIONS

allocation_id	*request_id*	*hall_id*	*slot_id*	booking_date	allocated_at

- request_id → BOOKING_REQUESTS (UNIQUE)
- hall_id → SEMINAR_HALLS
- slot_id → TIME_SLOTS
- UNIQUE (hall_id, booking_date, slot_id)

FREED_SLOTS new: our college rules

release_id	*request_id*	reason	freed_at

- request_id → BOOKING_REQUESTS (UNIQUE, ON DELETE CASCADE)
- rows are written by trigger trg_free_hall_on_cancel